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Quick Work
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M T W T F S S

Implementation of Subnetting in Cisco Packet Tracer.

Aim:

To implement Classless IP Subnetting and configure a network using Cisco Packet Tracer

Classless IP Subnetting.

* Subnetting divides a large network into Smaller network.
* Classless Subnetting allows subnets masks other than the default Class A, B or C masks

Network Topology

* Routers
* Switches
* PC's

Subnetting used

Given network - 192.168.1.0/24

To create Subnets with Sufficient host address, use:

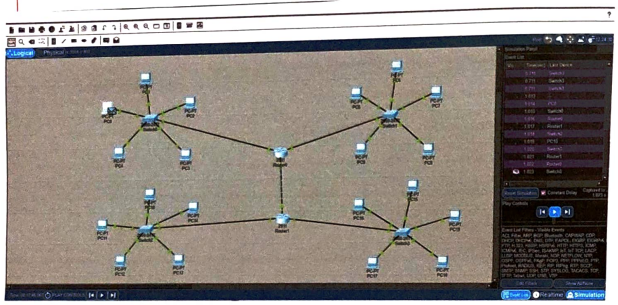
/27 = 255.255.255.224.

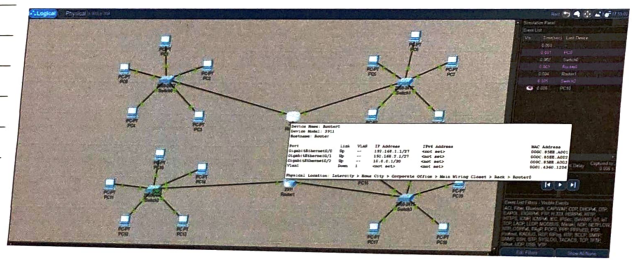
No: of Subnets = 8

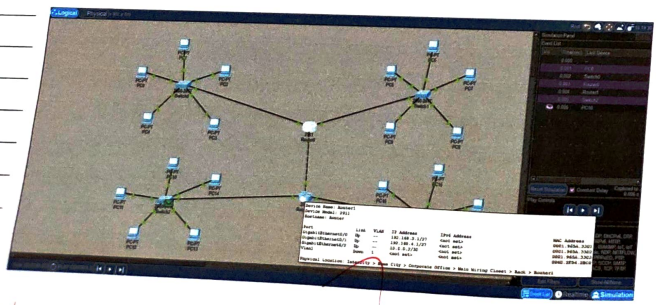
Block Size : 32

Ex: 192.168.1.0/27

192.168.1.32/27.







Procedure:

* Open Cisco Packet Tracer and create a new network topology.
* Add the required routers, switches & PC's
* Connect the devices using appropriate cables
* Divide the network 192.168.1.0/24 into smaller subnets using a /27 subnet mask.
* Configure the switch port in access mode
* Test connectivity between PC's and routers using the ping command.
* Set the appropriate default gateway for each PC.

Result:

Thus, the subnetted network was successfully configured and connectivity was verified using ping.